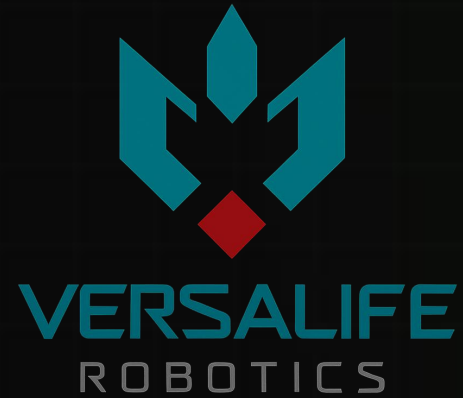




PRESENTS

AEROGUARD **SENTINEL**

Advanced Precision Airborne EOD Disruption Platform

PHASE 01

THE CHALLENGE

THE URBAN IED THREAT

COMPLEXITY IN CHAOS

Modern improvised threats are increasingly deployed in rubble-heavy, confined, or elevated urban environments. Disarming these devices requires surgical precision under extreme duress.

- ⚠ High-risk human exposure in the blast zone.
- ⚠ Minutes lost navigating through urban debris.



PHASE 02

CURRENT SOLUTIONS

UNMANNED GROUND VEHICLES (UGVS)



THE INDUSTRY STANDARD

Current EOD doctrine relies heavily on treaded ground robots. These machines are designed to carry heavy payloads and manipulate objects at ground level.

While robust, they operate on a 2D plane, restricted by the physics of the terrain they occupy.

| THE UGV MOBILITY GAP

-  **Terrain Barriers:** Ground robots are frequently blocked by collapsed stairs, steep rubble, or flooded corridors.
-  **Deployment Lag:** Navigating a ground robot to a target 100m away can take up to 15 minutes in complex debris.
-  **Visual Obstruction:** Ground-level sensors lack the high-angle perspective needed to inspect complex wiring or triggers.
-  **Mechanical Bulk:** Heavy chassis are difficult to transport manually to the frontline of a conflict zone.

PHASE 03

OUR SYSTEM

INTRODUCING AEROGUARD SENTINEL

THE AERIAL DISRUPTION ADVANTAGE

Versalife Robotics presents a paradigm shift in EOD. The Sentinel bypasses the ground entirely, delivering precision disruption payloads from the air.

By using high-torque robotic manipulation combined with aerospace precision, we neutralize threats in seconds, not hours.



| THE GROUND STATION

RTK + OPTICAL FLOW

Integrated RTK-GNSS and high-resolution Optical Flow sensors ensure the drone "locks" into space, even in GPS-denied urban canyons or under bridges.

ADVANCE CONTROLLER

Remote Controller Seamlessly connected with the drone system for precise navigation and low-latency control. Designed for reliable and accurate operation during critical IED defusing missions.



MILLIMETER-LEVEL CONTROL



3-AXIS ROBOTIC MANIPULATOR

The Sentinel's core is its high-precision 3-axis arm, specifically designed to carry and plant secondary charges onto localized threats.

This allows for surgical "Blow-in-Place" (BIP) operations with zero ground-contact risk.

TECHNICAL SPECIFICATIONS

Component	AeroGuard Sentinel v2.5	Performance Metric
GNSS System	Multi-Constellation RTK (L1/L2/L5)	2-3cm Positioning Accuracy
Visual Sensors	Down-facing Optical Flow	Active below 10m AGL
Manipulator	3-Axis Robotic Arm	500g Payload @ 30cm reach
Flight Stability	Triple-Redundant Isolated IMUs	< 0.01° Attitude Error
Flight Time	High-Energy LiPo Power Cell	10 Minutes with full payload

MISSION EFFICIENCY IMPACT



Versalife Robotics provides an 82% mission time reduction vs. ground-based units.

| SAFETY FIRST

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"The Sentinel v2.5 doesn't just improve efficiency; it fundamentally shifts the risk profile of EOD, ensuring our most valuable assets—our personnel—remain safe."

MISSION AND RESEARCH-DRIVEN INNOVATION

ABOUT VERSALIFE ROBOTICS

Driven by innovation and purpose, we are an ambitious startup on a mission to build systems that protect people across **medical, industrial, and defense** sectors.

GuardianEye is our first step toward a future where robotics work seamlessly — and safely — alongside humans.

📍 Ravet, Pune, Maharashtra

🔧 Specialized in Embedded Edge AI and Robotics.





VERSALIFE

ROBOTICS

Securing the future of tactical neutralization.

 versaliferobotics@gmail.com

 +91 8767305818

 Ravet, Pune, India - 412101